

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTEEPE405C - ADVANCE RENEWABLE ENERGY SOURCES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 Which of the following is a renewable source of energy?
 - a. Coal
 - b. Natural Gas
 - c. Solar Energy
 - d. Nuclear Energy
- Modern electronic controls in power systems help in
 - a. Increasing the carbon emissions
 - b. Making manual operation mandatory
 - c. Improving system stability and efficiency
 - d. Disconnecting renewable sources
- 5 The main challenge in commercializing fuel cells is
 - a. Lack of fuel
 - b. High cost and storage of hydrogen
 - c. High voltage output
 - d. Incompatibility with electronics
- 4 Proton Exchange Membrane Fuel Cells (PEMFC) operate typically at
 - a. Below 100°C
 - b. Around 300°C
 - c. 500–600°C
 - d. Above 1000°C
- 5 The instrument used to measure wind speed is called
 - a. Barometer
 - b. Anemometer
 - c. Hygrometer
 - d. Pyranometer
- In wind turbine site selection, which of the following is least important?
 - a. Wind intensity
 - b. Topography
 - c. Soil type
 - d. None of the above
- Which device is used to measure solar radiation?
 - a. Anemometer
 - b. Pyranometer
 - c. Photometer
 - d. Radiometer
- 8 The efficiency of a solar cell is defined as the ratio of
 - a. Output voltage to input current
 - b. Output power to input solar power
 - c. Input current to output voltage
 - d. Heat loss to solar input
- 9 Which of the following is a thermochemical biomass conversion process?
 - a. Anaerobic digestion
 - b. Alcohol fermentation
 - c. Gasification
 - d. Hydrolysis

- 10 In pyrolysis, biomass is decomposed
 a. In presence of large amounts of oxygen b. Without heat
 c. At high temperature in absence of oxygen d. Only in sunlight
- 11 SMES stands for
 a. Static Magnetic Energy Source
 b. Superconducting Magnetic Energy Storage
 c. Solar Magnetic Energy System
 d. System for Managing Energy Storage
- 12 Interconnection of alternative energy sources with the grid must follow
 a. Random user preferences
 b. Specific codes and standards
 c. Time-based tariff rules only
 d. Fuel cost criteria
- Q2. Solve the following.
- A) State and explain the different types of renewable energy sources. Also, discuss their advantages and disadvantages. 6
- B) With a neat diagram, explain the construction and working of a fuel cell. 6
- Q3. Solve the following.
- A) Explain the factors considered for appropriate site selection of wind farm. 6
- B) Draw and explain Wind Power Generation Curve and also explain the terms 6
- Cut-in speed (V_c)
 - Cut-out speed (V_s)
 - Rated speed (V_r)
 - Tip speed ratio
 - Power output
- Q4. Solve Any Two of the following.
- A) With neat diagrams, explain the solar flat plate collector and the solar concentrating collector. 6
- B) Draw and explain the electricity generation by photovoltaic effect. 6
- C) With neat diagrams, explain how solar thermal energy is used in solar cooking and drying systems. 6
- Q5. Solve Any Two of the following.
- Explain in detail the small-scale pyrolysis unit with the help of neat diagram. 6
- Compare Synchronous Generator and Induction Generator based on construction, working principle, and applications. 6
- Explain different types of biomass conversion process. 6
- Q6. Solve Any Two of the following.
- A) Explain the energy storage mechanism of an Ultra Capacitor. Compare it with a Lead-Acid Battery. 6
- B) Draw a neat diagram and explain the construction and working of a Lead-Acid Battery. 6
- C) With suitable examples, explain how different types of renewable energy sources are interconnected to the grid. 6